

PATRICK AFB, FL, USA

WMO: 747950

Lat: 28.233N

Lon: 80.600W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 12867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	9.3	-3.3	2.9	10.2	0.3	3.8	12.0	12.5	16.3	11.5	15.8	6.7	320	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	32.5	26.1	31.8	26.3	31.2	26.2	28.0	30.0	27.6	29.8	27.1	29.6	4.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	23.6	29.1	27.0	22.7	28.7	26.5	22.0	28.6	89.6	30.0	87.2	29.9	85.2	29.6	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.1	8.3	3.5	34.8	1.8	0.6	2.2	35.2	1.2	35.6	0.2	35.9	-1.1	36.3		
			1.1	28.2	2.0	1.1	-0.3	29.0	-1.5	29.6	-2.6	30.3	-4.0	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	17.4	19.6	20.7	23.8	25.6	27.4	28.2	27.9	25.6	22.0	19.7		
		DBStd	4.52	3.88	3.43	3.31	1.89	1.52	1.13	1.15	0.89	1.07	2.38	4.15		
		HDD10.0	6	3	1	0	0	0	0	0	0	0	0	2		
		HDD18.3	153	64	24	19	0	0	0	0	0	1	8	37		
		CDD10.0	5066	232	269	332	413	483	521	564	537	482	359	303		
		CDD18.3	2172	34	60	92	163	224	271	306	313	287	225	117		
		CDH23.3	18217	83	168	302	870	1708	2636	3369	3481	3081	1850	457		
		CDH26.7	5098	9	22	37	117	328	778	1192	1221	977	361	39		
(14)	Wind	WSAvg	4.5	4.7	4.4	4.9	4.9	4.7	4.1	3.6	3.8	4.1	5.0	4.9		
(15)	Precipitation	PrecAvg	1318	58	52	77	59	85	171	158	188	180	135	58		
		PrecMax	1770	154	168	297	147	257	363	297	610	361	290	206		
		PrecMin	962	0	1	0	16	17	3	55	37	88	19	6		
		PrecStd	244	35	36	73	33	58	81	65	110	73	79	44		
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	29.0	30.1	31.1	32.0	33.8	33.5	33.7	33.0	31.3	29.4		
			MCWB	21.9	21.7	22.1	23.1	23.7	25.1	26.0	26.5	26.2	25.2	24.6		
		2%	DB	25.3	26.8	27.6	29.0	30.0	31.9	32.3	32.1	31.6	30.2	27.9		
			MCWB	21.2	21.7	21.5	23.2	24.3	25.8	26.5	26.7	26.0	25.4	23.6		
		5%	DB	24.0	25.3	26.0	27.9	29.1	30.8	31.7	31.4	31.0	29.4	26.7		
			MCWB	20.6	21.6	21.6	23.0	24.1	25.9	26.7	26.6	25.9	25.1	22.9		
		10%	DB	22.6	24.1	25.1	27.0	28.5	30.1	31.0	30.8	30.4	28.6	25.8		
			MCWB	20.0	21.1	20.9	22.4	23.8	25.6	26.4	26.5	25.7	24.0	22.0		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	23.9	23.9	25.3	26.2	27.9	29.0	29.5	28.0	27.8	25.8		
			MCDB	25.3	26.0	26.8	27.9	28.2	29.8	30.6	31.4	30.1	28.8	27.7		
		2%	WB	22.4	23.0	23.1	24.7	25.5	27.1	28.0	28.2	27.3	26.8	25.1		
			MCDB	24.3	24.6	25.5	27.2	27.9	29.7	30.2	30.0	29.8	28.4	26.8		
		5%	WB	21.4	22.3	22.4	24.1	25.1	26.6	27.5	27.5	26.8	26.1	24.0		
			MCDB	23.0	24.1	25.0	26.5	27.7	29.3	30.0	29.8	29.5	28.1	26.0		
		10%	WB	20.6	21.6	21.7	23.6	24.6	26.2	26.9	27.1	26.4	25.4	22.9		
			MCDB	22.1	23.5	24.3	26.0	27.3	28.9	29.7	29.5	29.3	27.8	25.0		
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	7.1	7.1	6.3	5.9	5.9	5.8	5.6	5.3	5.3	5.9		
			MCDBR	8.0	7.7	7.8	7.1	6.4	7.1	6.4	6.2	5.8	5.2	5.7		
		5% WB	MCWBR	4.6	4.2	3.9	3.6	3.1	3.3	3.0	3.3	2.7	2.7	3.1		
			MCWBR	7.2	6.8	7.0	6.4	6.0	6.1	5.9	5.6	5.5	4.9	5.2		
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.360	0.375	0.397	0.428	0.455	0.473	0.462	0.426	0.402	0.366	0.354		
		taud	2.537	2.514	2.460	2.377	2.331	2.330	2.280	2.328	2.419	2.453	2.508	2.543		
		Ebn at Noon	897	913	916	900	867	838	823	831	854	856	868	872		
		Edh at Noon	89	98	110	123	129	128	135	128	113	103	91	85		
(44)	All-Sky Solar Radiation	RadAvg	3.38	4.10	5.14	6.08	6.26	5.91	5.91	5.62	4.92	4.37	3.56	3.06		
		RadStd	0.24	0.24	0.33	0.24	0.44	0.55	0.31	0.34	0.30	0.28	0.19	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-40	+153	+112

Nomenclature: See separate page